Repetition, Information Preservation, and Decision Reliability in Hierarchical Language-Model Agents

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Abstract. Repeated reports are common in language-model agent systems, but repeating a finding can both preserve it during communication and change how a decision maker uses it. We study these effects with a deterministic benchmark for selecting the best feasible candidate schedule, alongside a separate rule-based workflow task. Frozen worker reports isolate repeated presentation from fresh computation and new evidence. Paired worlds test correct responses to consequential edits and correct stability under harmless changes. Flat aggregation is compared with two intermediate managers whose summaries form the root input. Completed pilot and validation evaluations cover 24 and 36 independent base problems, respectively. On hierarchical scheduling, exact copies improve Qwen3 pilot accuracy from 81.25% to 94.79%, but validation effects are -3.47 percentage points for Qwen3 and +1.39 for Phi-4, with both cluster intervals spanning zero. An audit of 6,336 manager summaries locates numerical objectives becoming Boolean values in raw generated summaries, with no subsequent parser or transport changes. This loss does not always make the decision ambiguous, and correct delivery does not guarantee a correct choice. The results motivate separate evaluation of representation fidelity, decision sufficiency, and decision accuracy. Main held-out and intervention results remain pending in this working draft. Conclusions are limited to the evaluated models, tasks, and structured communication interface.

Keywords: language-model agents, hierarchical aggregation, repetition, information preservation, behavioral evaluation

1 Introduction

A language-model manager may receive several reports that say nearly the same thing. Some are copies of an existing observation. Others add a constraint, refer to a different machine, or correct an earlier value. Reliable coordination requires the manager to preserve these distinctions while using all the information needed for its decision. Repetition creates a useful tension. It may help a fact survive summarization, yet it can also give a limited claim more influence than its relevance warrants.

Consider a team choosing among proposed schedules. One worker reports completion times, another checks machine capacity, and another checks dependencies. Repeating that one plan is fast does not establish that it is feasible. Equally, removing every similar sentence could discard the one capacity failure that changes the answer. These are integration problems. The reports answer complementary questions, so a majority vote over their favorable or unfavorable conclusions is not a valid decision rule.

Research on repeated evidence already distinguishes sources from restatements [1, 2]. Research on distributed information shows that agent teams can fail to discover or use facts held by their members [3]. Our focus is the transition between these settings. All required worker findings are initially delivered, but a hierarchical decision maker receives summaries rather than the original reports. This lets us examine whether a failure occurs during communication or after the relevant information has arrived.

We investigate three questions. First, how does repeating a report affect decision accuracy in flat and hierarchical organizations? Second, where do complete propositions lose their value or scope during aggregation? Third, when does such a loss actually prevent a justified decision? The distinction in the third question matters. A missing completion time is important when several feasible plans compete, but may be irrelevant when only one plan is feasible.

The study provides a deterministic benchmark for candidate selection, paired worlds with exactly checked changes, and a replay procedure that varies report presentation without adding evidence. It also provides a trace audit that follows assertions from the original report to the final model input, and an analysis of the decisions supported by that input. These components make it possible to separate representation fidelity, decision sufficiency, and final accuracy.

The completed development evidence illustrates this distinction. A substantial hierarchical pilot benefit did not consistently carry over to validation. Numerical objectives sometimes become Boolean values in raw manager summaries, but repetition does not reliably prevent this loss. Some decisions are also wrong despite correctly delivered information. Together, these findings locate different reliability problems without implying a universal rule for repeating reports.

2 Related Work

Repeated evidence and additional computation. Gao et al. [1] measure how aggregators count restated observations in controlled logs, web documents, and agent-written communication. Their experiments already include summaries and a hierarchy. In their agent-team setting, the aggregator sees the roster and full log. Our main hierarchy instead removes raw reports from the root's input and requires it to combine mandatory constraints with a numerical objective. This is a narrower distinction than claiming that evidence double counting or hierarchical communication is new. Bara [2] formalizes why report multiplicity and evidence multiplicity differ, including the possibility of correlated extraction errors.

Repetition can also be beneficial. Leviathan et al. [4] evaluate repetition of the whole input prompt in non-reasoning models. We selectively repeat a report and track what survives a summary boundary. Neither intervention is the same as self-consistency, which samples additional reasoning paths and aggregates their answers [5]. Fresh verification is additional computation. Exact copies are additional presentations of an existing result. Neither necessarily supplies a new independent observation.

Agent organization and distributed information. Multiagent debate can improve answers by exchanging separately generated responses [6], but its benefits depend on the

protocol and comparison budget [7]. The current version of the scaling study by Kim et al. [8] examines how architectures interact with task structure. We use its third preprint version and do not combine findings from earlier versions. HiddenBench [3] tests information distributed across agents, including failures to elicit unshared facts and a structured communication intervention. Our replay begins after the essential reports have been supplied. This isolates later aggregation from the problem of discovering missing evidence. The MAST taxonomy [9] places such failures within a broader range of coordination problems. Hi-Agreement [10] studies fact recovery in the presence of misinformation. It motivates keeping incorrect testimony separate from a legitimate change in the underlying world.

Faithfulness, scope, and presentation. Summarization research distinguishes a summary's faithfulness to its source from its factual correctness [11]. We add a task-specific question: does the delivered information determine the required choice? Behavioral testing with CheckList [12] motivates both meaning-preserving and consequential changes. PAWS [13] demonstrates why high word overlap does not establish semantic equivalence. Position effects in long contexts [14] and metadata effects in conflicting retrieved evidence [15] provide alternative explanations for apparent repetition effects. They motivate the study's separate ordering, length, scope, and provenance controls.

Prior conclusions and role labels are different interventions. Intrinsic self-correction can fail without external feedback [16], while identity labels [17] and authority roles [18] can influence multiagent debate or evaluation. These studies motivate separate probes of feedback and relayed reports. A changed answer alone does not reveal a model's belief about source independence or authority.

3 Task Setting and a Worked Example

3.1 Choosing Among Supplied Schedules

The primary task is to select the feasible candidate with the smallest makespan, where makespan is the latest job completion time. The model chooses from a supplied set of plans. It does not search for a globally optimal schedule.

Jobs have integer durations and resource demands. They run without interruption on half-open intervals $[s_j, s_j + d_j)$, so a job finishing at time 2 does not overlap a job starting at time 2. For resource r, total simultaneous demand must satisfy

$$\sum_{j : s_j \le t < s_j + d_j} q_{jr} \le C_r, \tag{1}$$

where C_r is capacity and q_{jr} is job j's demand. A precedence edge from u to v requires $s_u + d_u \le s_v$. Releases, deadlines, and resource downtime add mandatory constraints when present.

Write $F(c)$ for the conjunction of mandatory constraints and $M(c)$ for the makespan of candidate c. The choice is

$$c^* = \arg\min_{c : F(c) = 1} \big(M(c), \pi(c)\big), \tag{2}$$

Table 1. Illustrative plans. All intervals are half-open and times use the same unit.

Plan	A on M	B on M	C on N	Makespan
P	$[0,2)$	$[0,2)$	$[2,3)$	3
Q	$[0,2)$	$[2,4)$	$[4,5)$	5

where the public priority rank $\pi(c)$ resolves ties lexicographically. The answer is `NONE` if no supplied candidate is feasible. An attractive objective cannot compensate for a failed mandatory constraint.

3.2 An Illustrative Capacity Change

Consider jobs A and B, each requiring one unit of machine M for two time units. Job C requires one unit of a different machine N for one time unit and follows both A and B. Machine N has capacity one. Table 1 gives two candidate plans. This is an illustrative example checked with both benchmark verifiers, not an observed model error.

With machine M's capacity equal to one, P is infeasible because A and B overlap. Q is feasible and is the correct choice. Increasing M's capacity to two makes P feasible, so P becomes correct. Precedence and makespan reports remain unchanged. Only the resource finding for P changes.

An objective worker contributes the statement that P's makespan is 3. A resource worker contributes the separate statement that its peak demand on M is 2 and whether that meets the applicable capacity. Four exact presentations of the objective report still describe one computed result. They cannot establish that P is feasible when capacity is one.

The same example explains decision sufficiency. At capacity one, the feasibility findings alone identify Q, even if both numerical objectives disappear. At capacity two, both plans are feasible. If their objective values are missing, the remaining reports do not determine which is best. A representation error can therefore be irrelevant to one decision and decisive for another.

4 Experimental Design

4.1 Instances, Reports, and Paired Changes

The generator builds small schedules from valid structures and controlled alterations. The evaluated schedules use three candidates and two or three resource types. Candidate identifiers, ordering, and tie priorities are randomized without consulting a model. An event-sweep checker evaluates each candidate, and a separately implemented integer-time occupancy checker cross-checks its feasibility and objective.

Each independent base problem has an original and a modified world. The pair records the changed fields and both correct answers. A consequential edit changes the correct choice. A harmless edit leaves it unchanged. Scheduling pairs cover capacity, resource identity, and precedence changes. A separate deterministic workflow family

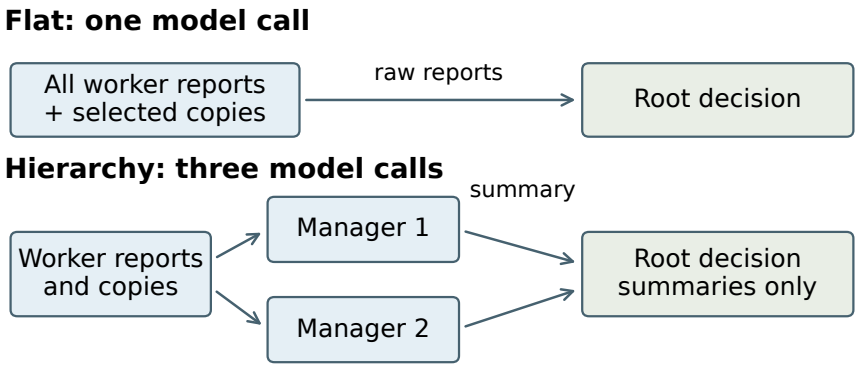


Fig. 1. Flat and hierarchical replay. Copies add presentations, not sources or worker executions.

covers quantification, conditional rules, and explicit version selection. Workflow results are reported separately because these edits have a different task structure.

Workers' findings are typed assertions rendered with fixed templates. The replay therefore starts with known reports rather than model-generated worker errors. Each claim identifies its task, candidate, entity, predicate, value, units, applicability, and version. It also records sources, parent reports, and the verification execution. Two claims are equivalent only when their meanings and scopes agree. A different candidate, resource, condition, or effective version can make similar wording express a different proposition.

The changed-world replay regenerates affected reports consistently. Separate corrupted-report controls keep the world fixed and introduce a false assertion with visible evidence that can resolve it. A contradiction is not automatically a valid objection. Likewise, a newer report has authority only when the task defines an update rule. These controls must not be pooled with the all-correct-report replay as if they were the same intervention.

4.2 Repetition and Organizational Structure

The baseline presents every required result once. Exact repetition adds three copies of one selected report. Paraphrasing adds three template-rendered restatements of the same claim, retaining its value, scope, and ancestry. Thus the selected report has one or four total presentations. Essential reports are never removed to make room. A prespecified rotation selects the target, preferring a report unchanged across paired worlds. This does not guarantee that the selected report is decisive or favors the correct candidate.

Figure 1 shows the access difference. A flat root reads the complete report set in one call. In the hierarchy, two managers receive fixed partitions, summarize their reports, and pass the summaries to a root. The root receives a registry of required claim scopes but no raw worker reports. This registry contains descriptions and ancestry, not evalua-

tor truth values. Copies initially stay in their report's branch. Every policy receives the same available information within an arm.

Three policies vary the aggregation instructions. Ordinary synthesis asks for a decision and its explicit supporting findings. A coverage checklist asks the model to account for each mandatory predicate before comparing objectives. Scope-aware consolidation additionally asks it to merge equivalent claims while preserving different entities, versions, conditions, and unresolved conflicts. Even ordinary synthesis receives the full decision rule. The common output contract also asks the root for a feasibility and objective assessment of each candidate. This is a substantial baseline scaffold, so the comparison concerns incremental instructions within a structured interface.

Managers emit keyed assertions and unresolved keys. Roots additionally emit a choice, candidate assessments, and probabilities. The study scores these public outputs, not private reasoning. The JSON decoder enforces syntax and field structure. It does not enforce all semantic relationships between a scope key and its value. Section 6 examines the consequences of that distinction.

4.3 Models, Stages, and Experimental Integrity

We use Qwen3-14B [19] with thinking disabled and Phi-4 [20]. Both run sequentially in bfloat16 without quantization on one NVIDIA RTX 6000 Ada Generation GPU. Sampling uses temperature 0.3 and top-p 0.95. The replay has a 12,288-token context limit, a 2,304-token output cap, and batches of up to 32 requests. Native chat templates and exact model revisions are fixed. Appendix A gives implementation details.

Table 2. Completed replay stages. Bases are independent problems, shared across models.

Stage	Model	All bases	Scheduling bases	Trajectories
Pilot	Qwen3-14B	24	12	1,152
Validation	Qwen3-14B	36	18	2,592
Validation	Phi-4	same 36	same 18	2,592

Table 2 separates independent problems from repeated trajectories. The pilot uses ordinary synthesis and the checklist. Validation adds the scope-aware policy and a second model. To compare stages fairly, the main pilot-versus-validation estimates average only the two shared policies. Other validation comparisons explicitly average all three. Every cell has two stochastic repetitions, with independent cache identities. All variants of a base problem stay in the same split. Validation changes parameter ranges and rendering conditions, while the main test additionally reserves graph patterns and templates. New seeds alone are not treated as sufficient separation.

The original hypotheses asked whether repetition changes error, reduces correct responses to consequential edits, interacts with hierarchy, and is better handled by scope-aware consolidation than a checklist. After inspecting the pilot, a dated amendment broadened the mechanism questions to include beneficial information preservation.

The original hypotheses and runs remain intact. Reanalyses motivated by the pilot are exploratory. Fresh confirmation and mechanism interventions remain separate stages, rather than being combined with development into a larger apparent sample.

Immutable records identify requests, responses, corpus, prompts, model, and inference settings. Resuming a run reuses completed calls, including malformed answers, which count as errors. Analyses must be complete before their estimates enter the paper.

4.4 Outcomes and Uncertainty

Exact agreement with the executable answer is the primary outcome. We also distinguish infeasible selections, feasible but suboptimal selections, and incorrect `NONE` or `INSUFFICIENT` responses. Paired-world success requires both answers to be correct. On consequential pairs this measures correct responsiveness, and on harmless pairs it measures correct stability. An answer flip alone satisfies neither criterion.

For base problem b, let $\bar{Y}_{b,a,r}$ be accuracy averaged over the specified worlds, policies, and stochastic repetitions in organization a and presentation condition r. We report the accuracy effect

$$\Delta_{a,r} = \frac{1}{B}\sum_{b=1}^{B}(\bar{Y}_{b,a,r} - \bar{Y}_{b,a,\text{single}}). \tag{3}$$

Positive values favor repetition. The original error-based repetition-harm contrast is $-\Delta_{a,r}$. We estimate the hierarchy interaction directly as $\Delta_{\text{hierarchical},r} - \Delta_{\text{flat},r}$. A lower overall hierarchical accuracy does not by itself show that hierarchy amplifies repetition.

Uncertainty uses whole base-problem clusters [21]. Each bootstrap draw retains the associated worlds, arms, policies, and repetitions. The displayed exploratory estimates use 10,000 draws stratified by mutation family and whether the correct decision changes. Models are reported separately. Intervals are percentile 95% intervals, not multiplicity-adjusted tests. We make no confirmatory significance claims from these development analyses. Few clusters limit precision, and a zero-width interval at a sample ceiling is not evidence of perfect population performance.

5 Results

5.1 Evidence Available in This Draft

This draft reports the completed pilot, exploratory validation reanalysis, and trace audits. The pilot contains 24 independent base pairs and 1,152 trajectories. Validation contains 36 additional independent pairs shared across two models and 5,184 trajectories. The smaller development controls and live-worker checks are diagnostic rather than confirmation. **Pending results.** Results remain pending for the main held-out replay, fresh objective confirmation, the root-only diagnostic, the objective-preservation replay, and the remaining frozen supporting evaluations. No partial held-out outcomes enter the results below.

5.2 A Pilot Benefit and a More Mixed Validation Result

For hierarchical scheduling, the pilot compares 96 decisions per presentation condition from 12 independent bases. Each base contributes two worlds, two policies, and two stochastic repetitions. Validation contributes 144 decisions per condition and model from 18 bases under the same two-policy average.

Table 3. Hierarchical scheduling, two shared policies. Changes use paired base-cluster 95% intervals.

Stage / model	Repeat	Single %	Repeated %	Change, pp [95% CI]
Pilot Qwen3	exact	81.25	94.79	+13.54 [+4.17, +22.92]
Pilot Qwen3	paraphrase	81.25	91.67	+10.42 [+1.04, +19.79]
Validation Qwen3	exact	79.86	76.39	-3.47 [-10.42, +2.78]
Validation Qwen3	paraphrase	79.86	78.47	-1.39 [-5.56, +2.78]
Validation Phi-4	exact	72.22	73.61	+1.39 [-4.17, +6.25]
Validation Phi-4	paraphrase	72.22	72.92	+0.69 [-4.17, +4.86]

Table 4. Paired decision transitions for Table 3. Unchanged includes correct and incorrect choices.

Stage / model	Repeat	Bases	Corrections	Regressions	Unchanged
Pilot Qwen3	exact	12	14	1	81
Pilot Qwen3	paraphrase	12	12	2	82
Validation Qwen3	exact	18	7	12	125
Validation Qwen3	paraphrase	18	7	9	128
Validation Phi-4	exact	18	8	6	130
Validation Phi-4	paraphrase	18	8	7	129

Table 4 separates corrections from regressions. Exact repetition raises pilot accuracy from 81.25% to 94.79%, a gain of 13.54 percentage points (95% interval 4.17 to 22.92). Paraphrases raise it to 91.67%, a gain of 10.42 points (1.04 to 19.79). In validation, the corresponding exact-copy effects are -3.47 points for Qwen3 and +1.39 for Phi-4. Both validation intervals span zero. No parsing or truncation change explains these pilot corrections.

Pilot exact-copy effects are positive on six bases, negative on one, and zero on five. One base contributes seven of the 14 corrections. Leaving it out descriptively lowers the gain to 6.82 points, but it remains in every primary estimate. Across all single-base omissions, the pilot exact-copy gain ranges from 6.82 to 15.91 points. Figure 4 displays every base rather than selecting favorable examples.

The useful comparison in Figure 2 is between the hierarchical pilot estimates and the corresponding validation estimates, then between organizations within validation.

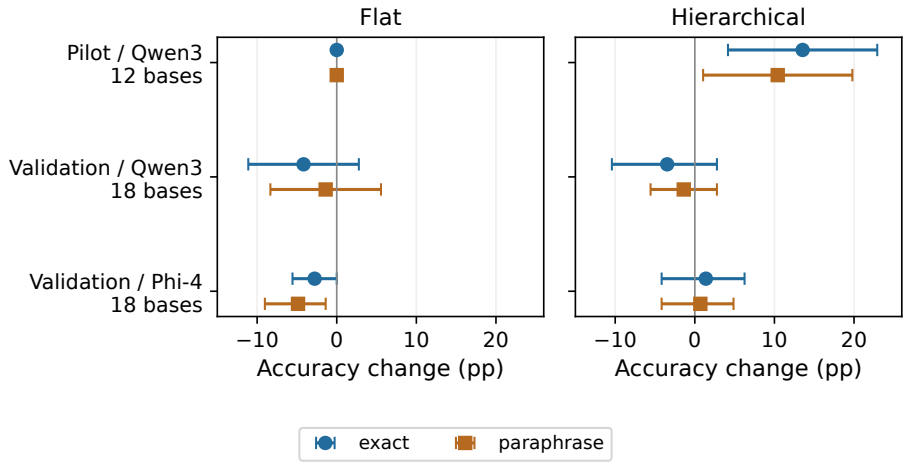


Fig. 2. Scheduling accuracy effects, repeated minus single. Bars are 95% base-cluster intervals.

The pilot pattern does not establish a general benefit of repetition. Nor do the validation intervals establish equivalence to no repetition. They allow small benefits and harms under the tested conditions.

The flat pilot scheduling arms all achieve 100% on these 12 bases, leaving no observed room for improvement. Validation flat accuracy under the shared policies changes from 94.44% to 90.28% with exact copies for Qwen3, and from 93.75% to 90.97% for Phi-4. The exact-copy hierarchy interactions are Qwen3-14B +0.69 [-6.25, +7.64], and Phi-4 +4.17 [-2.78, +11.11] percentage points. These direct differences of effects do not establish that hierarchy consistently amplifies or suppresses repetition.

The direct hierarchical exact-minus-paraphrase contrasts in validation are Qwen3-14B -2.08 [-9.72, +4.86], and Phi-4 +0.69 [-4.17, +5.56] percentage points. Separate significance judgments for the two repetition arms would not test this difference.

The validation distribution differs from the pilot in task parameters and case composition, including ties and no-feasible-candidate cases. This limits the interpretation of the stage comparison. It demonstrates that a promising development average need not transfer to the next evaluation set, but does not identify which changed feature explains the difference.

5.3 Responsiveness to Consequential Changes

Paired worlds ask a stricter question than average single-world accuracy. A model must be right before and after an edit. Table 5 separates consequential edits from harmless controls and includes all three validation policies. It therefore has a different policy denominator from Table 3.

Qwen3-14B: the consequential-minus-harmless difference in repetition effects is -3.70 [-14.81, +7.41] percentage points.

Table 5. Both worlds correct in hierarchical validation. Each row has nine bases and 54 paired trials per arm.

Model	Edit	Single %	Exact %	Change, pp [95% CI]
Qwen3-14B	harmless	68.52	64.81	-3.70 [-12.96, +3.70]
Qwen3-14B	consequential	79.63	72.22	-7.41 [-14.81, -1.85]
Phi-4	harmless	74.07	75.93	+1.85 [-3.70, +7.41]
Phi-4	consequential	46.30	42.59	-3.70 [-20.37, +9.26]

Phi-4: the consequential-minus-harmless difference in repetition effects is -5.56 [-24.07, +9.26] percentage points.

Qwen3 has a lower paired-world success rate with exact copies on consequential edits, but the direct contrast with harmless edits remains uncertain. The corresponding Phi-4 contrast is also imprecise. Neither comparison establishes a general selective loss of consequential differences.

These are changed-world tests with consistently regenerated reports. They do not by themselves measure whether two similar statements within one input are incorrectly merged. The frozen overlap and exception controls address that separate question. In particular, a decrease in paired-world success should not be renamed exception loss without evidence that the exception disappeared or changed scope in the delivered summary.

5.4 Policy and Resource Comparisons

The ordinary policy, checklist, and scope-aware policy share the same explicit decision rule and response schema. Differences between them are useful tests of additional instructions, but not a comparison with an unstructured conversation. Appendix B gives policy-specific validation accuracy, and Figure 5 relates accuracy to measured token use. No policy is credited with privileged knowledge of the correct answer.

In validation, more elaborate consolidation instructions do not produce a uniform ordering across models, organizations, and presentation conditions. The full policy table retains improvements and regressions. The shared structured response contract and the small number of independent bases limit claims about one policy being generally preferable.

6 Information Preservation and Decision Sufficiency

6.1 Locating the Numerical Transformation

The audit follows the original typed report, actual manager input, raw generated summary, parsed assertions, serialized summary, and actual root input. It covers 6,336 manager summaries from 3,168 hierarchical trajectories across the completed pilot and validation. These trajectories come from 60 independent base problems, with validation problems shared across models.

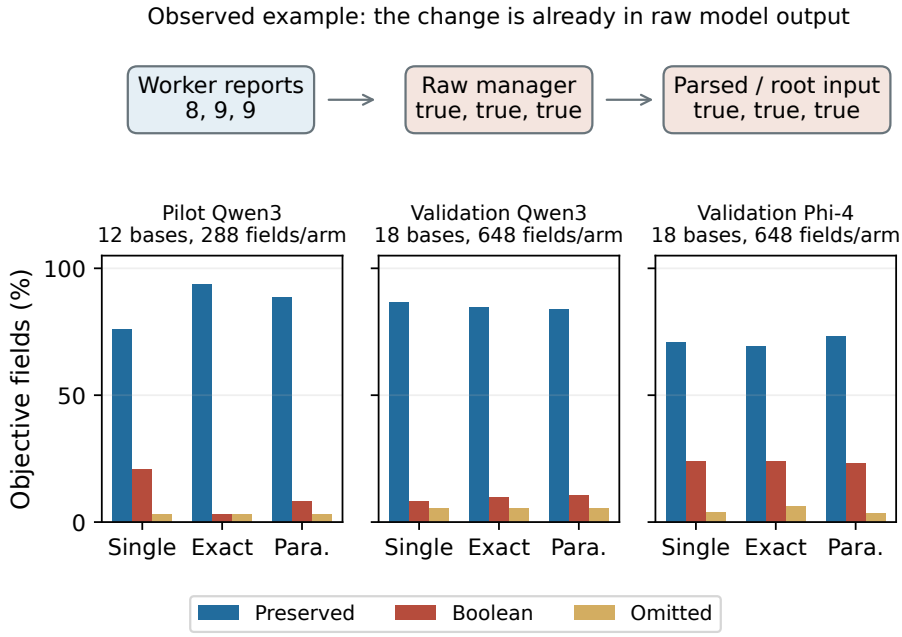


Fig. 3. Objective-field fidelity and an observed transformation. Pilot uses two policies, validation three. Fields within bases are dependent.

Numerical objectives becoming Boolean `true` first appear in the raw model-generated manager summaries. The audit found no value changes introduced by parsing, serialization, or root-input assembly, and no disagreement between raw final choices and their primary scores. It also found no truncated manager outputs. Within the audited pipeline, this is observed model behavior under the specified interface, not a downstream transformation defect.

The instructions explicitly require numerical objectives, but the assertion schema permits either a Boolean or a number at every scope key. Its illustrative assertion uses `true`. Thus a syntactically valid answer can replace a makespan with a truth indicator. The interface permits this semantic error. The audit does not establish that the example caused it, and the frequency need not generalize to a schema that enforces a numeric type for objective keys.

Figure 3 separates preserved numerical fields from Boolean replacements and omissions. Table 6 counts each candidate objective once per trajectory, regardless of copies. The pilot reduction in Boolean replacements is not repeated in validation: Qwen3 has 53 such fields without repetition and 64 with exact copies, while Phi-4 has 156 in both arms. Every numerical value emitted at a known objective key in the audited summaries matches its source. This does not count omitted, unresolved, or ill-typed values as preserved.

Table 6. Boolean objective fields in hierarchical scheduling. Pilot uses two policies, validation three.

Group	Single	Exact	Paraphrase
Pilot Qwen3	60/288	9/288	24/288
Validation Qwen3	53/648	64/648	69/648
Validation Phi-4	156/648	156/648	150/648

Using the same two policies as Table 3, exact-copy effects on objective preservation are Pilot Qwen3 +17.71 [+3.12, +32.29], Validation Qwen3 -4.86 [-12.50, +0.69], Validation Phi-4 -1.39 [-6.25, +4.86] percentage points. These are retention effects, not accuracy effects.

The influential pilot pair offers a concrete illustration of the observed path. All three candidates are feasible and their objectives are 8, 9, and 9. All eight unrepeated hierarchical contexts replace those numbers with `true`, while all eight exact-copy contexts preserve them. The repeated objective belongs to a candidate with value 9, rather than the correct candidate with value 8. Seven of the unrepeated decisions are wrong and all eight repeated decisions are correct. This suggests a useful numerical-preservation hypothesis, but does not isolate it from changes in attention, summarization, or final integration.

6.2 What the Delivered Information Supports

Representation fidelity asks whether a complete proposition survives with the same candidate, value, units, condition, and version. Decision sufficiency asks whether the visible information determines a choice. World accuracy asks whether the chosen answer matches the executable task. These outcomes can disagree. A lucky correct choice after a necessary fact is lost is still correct at the world level, but is not evidence of successful integration of that fact.

The visible-information scorer first identifies missing values, invalid types, and conflicting assertions. It does not replace them with hidden ground truth. For a separate decision-support calculation, it holds uncontested visible assertions fixed and varies unknown or disputed predicates and objectives over admissible values in the report interface. Boolean predicates can be true or false, and objectives are nonnegative integers. A decision is determined only if every admissible completion yields the same answer under the public tie rule. Being possible under one completion is not enough.

This calculation is an abstraction of the report interface, not an enumeration of all underlying job schedules. It can conservatively leave a decision unresolved that additional domain relationships would determine. It also does not certify that uncontested assertions are true in the world. Contradictions and invalid representations remain explicitly labeled even when they concern an irrelevant field and the remaining information already fixes the choice.

Pilot counts use two policies and validation counts use all three. The invalid-field column is a subset of contexts with a unique supported choice, not a repair of their representation. In validation, Qwen3 violates 24 of 195 uniquely supported choices and

Table 7. Unrepeated hierarchical scheduling. Violations use the uniquely supported contexts as denominator.

Group	Trajectories	Unique choice	With invalid fields	Violations
Pilot Qwen3	96	75	4	2/75
Validation Qwen3	216	195	10	24/195
Validation Phi-4	216	133	13	15/133

Phi-4 violates 15 of 133. These are conditional descriptive counts, not causal mediation estimates.

Table 7 separates communication from decision application. When the supported answer is unique but the root chooses otherwise, an objective repair alone need not solve the failure. Conversely, a numerical omission need not change the supported answer when mandatory findings already identify the only feasible candidate. This is why raw fidelity counts and final accuracy should not substitute for one another.

6.3 Observable Failure Locations

The trace analysis permits several labels per trajectory. A required proposition can be omitted, emitted with the wrong type or value, or linked to an incompatible scope. At the decision stage, a selected candidate can violate a delivered mandatory failure, or be feasible but worse than another feasible candidate. A correct choice can also be accompanied by erroneous supporting assertions. Contract failure is a separate operational label.

Keyed assertions and executable outcomes support deterministic scoring of these cases. An incompatible citation reveals a scope mismatch, but not an internal transfer of belief. Shared wording alone cannot establish an incorrect semantic merge. The categories describe public outputs, not a faithful record of internal reasoning.

Table 8. Unrepeated hierarchical validation failure labels. Each model has 216 trajectories from 18 bases.

Model	Omission	Distortion	FAIL ignored	Suboptimal	Contract
Qwen3-14B	12	8	9	6	0
Phi-4	7	32	15	23	0

Table 8 separates communication and application failures. Omission and distortion concern propositions whose individual removal from the complete report set makes the answer undetermined. FAIL ignored means an infeasible selected candidate has a correctly delivered mandatory failure. Suboptimal means a feasible selection has a strictly worse objective than the best feasible candidate. Labels can overlap and should not be summed.

7 Discussion and Limitations

7.1 Designing for Facts and Decisions

Repeating a report and preserving its contents are different operations. The validation evidence does not justify relying on repetition to preserve numerical values consistently. A schema with scope-dependent value types is a plausible engineering response, whose effect on accuracy requires controlled evaluation.

Preservation and correct application are also separate goals. Evaluation should check both the communicated propositions and the decision rule they support. This matters for complementary constraints, where several favorable reports cannot cancel one mandatory failure.

Reporting case-level effects reveals concentration that an average can hide. The influential pair remains in the pilot analysis. Validation refines the question from whether repetition helps to which information and interfaces make it useful, without identifying a cause of non-generalization.

7.2 Scope of the Evidence

These synthetic tasks measure controlled candidate selection, not deployment failure rates. Scheduling and the workflow share a decision structure, limiting claims of transfer across tasks.

Template reports isolate aggregation while omitting real workers' errors. Fresh executions using the same model and source can share errors, so additional computation need not supply independent evidence.

The hierarchy changes access and call count, requiring bypass and call-matched controls to separate them. Deeper hierarchies and feedback dynamics remain outside this replay.

The JSON contract and two model configurations constrain generalization. Allocated batch time describes this implementation, not individual latency, GPU kernel time, or monetary cost.

The audit is observational and cannot identify causal mediation. Deterministic preservation is an idealized control, not evidence that an autonomous summarizer achieves the same fidelity. Small targeted diagnostics require fresh confirmation before supporting broader mechanism claims.

8 Conclusion

Controlled replay separates repeated presentation from additional evidence. A large hierarchical pilot benefit is followed by smaller, mixed validation effects. Numerical objectives sometimes become Boolean assertions in manager output, but repetition does not reliably prevent this loss. Delivered information can remain sufficient despite a representation error, and a root can decide incorrectly despite a uniquely supported choice. Separating fidelity, sufficiency, and accuracy gives a more informative account of reliability. It also identifies a practical goal for future interventions. Preserve the required propositions, then verify that the decision follows from them.

A Reproducibility and Output Semantics

All paper tables and figures are generated from completed analysis records linked to immutable raw outputs. The reproducibility package contains the generators, both scheduling verifiers, report templates, prompt builders, frozen manifests, and base-cluster analyses. The anonymous manuscript omits identifying repository links. The source file contains the complete paper and bibliography.

The model and tokenizer revisions are identical within each family. For Qwen3-14B the revision is

```
40c069824f4251a91eefaf281ebe4c544efd3e18
```

For Phi-4 it is

```
2db69c1c3e91a05d2c64a3185acfbaf36f744e25
```

The licenses are Apache 2.0 and MIT, respectively. The environment uses vLLM 0.10.2, PyTorch 2.8.0 with CUDA 12.8, and Transformers 4.55.4. The GPU is an RTX 6000 Ada Generation with 49,140 MiB reported memory and driver 595.91.07. Prefix caching is disabled. Native chat-template token IDs are submitted directly, and template-inserted dates are pinned to 16 October 2026. Sampling seeds are derived from the base identifier, stochastic repetition, and stage. The seed scheme is shared across paired arms, but their request hashes remain distinct.

A claim registry contains task, candidate, entity, predicate, units, applicability, and version fields. Report ancestry separately records original sources, parent reports, operation type, and execution identifiers. The manager emits assertions of the form

`{"key": "scope key", "value": 8, "reports": ["report ID"]}`.

Its response also lists unresolved keys, with no final choice. The root adds a candidate assessment, choice, and probability distribution. A Boolean value at an objective key is not accepted as the number one by the evaluator. Valid integral floating-point objectives, such as 8.0, remain numerical values.

Full-scope objective-field counts use one required candidate objective per trajectory, regardless of how many copies were presented. A field is preserved only when its correct scoped value survives without an unresolved conflict. Omission, wrong type, conflicting values, and citation mismatch have separate indicators and can overlap. Summary counts and trajectory counts are not independent sample sizes. The independent unit for uncertainty remains the base problem.

B Policy and Case-Level Results

Table 9 reports each policy separately, avoiding an attribution of pooled performance to provenance instructions alone. The scope-aware policy adds instructions as well as consolidation guidance. A similar accuracy is not an equivalence finding, and a difference cannot be assigned solely to ancestry handling without the corresponding ablation.

Figure 4 exposes concentration that an average can hide. Each bar averages both worlds, two policies, and two stochastic repetitions within one base. The pilot contains a

Table 9. Validation scheduling accuracy (percent). Each cell has 18 bases and 72 trajectories.

Model	Organization	Policy	Single	Exact	Paraphrase
Qwen3-14B	flat	ordinary	95.83	91.67	91.67
Qwen3-14B	flat	checklist	93.06	88.89	94.44
Qwen3-14B	flat	scope	88.89	93.06	90.28
Qwen3-14B	hierarchy	ordinary	84.72	76.39	76.39
Qwen3-14B	hierarchy	checklist	75.00	76.39	80.56
Qwen3-14B	hierarchy	scope	80.56	84.72	79.17
Phi-4	flat	ordinary	93.06	91.67	91.67
Phi-4	flat	checklist	94.44	90.28	86.11
Phi-4	flat	scope	86.11	80.56	83.33
Phi-4	hierarchy	ordinary	76.39	76.39	77.78
Phi-4	hierarchy	checklist	68.06	70.83	68.06
Phi-4	hierarchy	scope	63.89	65.28	62.50

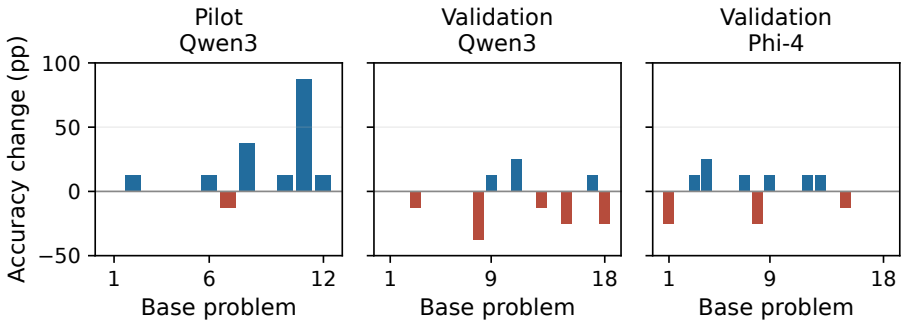


Fig. 4. Exact-copy effects for every scheduling base, in fixed identifier order. No cases are excluded.

large positive case, while the validation panels contain both corrections and regressions. The case ordering is fixed by identifier, not effect size. These plots are descriptive and do not select a subpopulation for subsequent confirmation.

Figure 5 includes every repetition arm under the three-policy validation average. Hierarchy consumes more tokens per decision and has lower accuracy than the corresponding flat organization in these data. This descriptive comparison combines access, summarization, and call-count changes. For unrepeated hierarchy, allocated batch seconds per correct decision are Qwen3-14B 10.98 and Phi-4 10.62. These values sum the recorded share of inference batch time and divide by correct decisions. They exclude model-loading overhead and are not per-request latency or monetary cost.

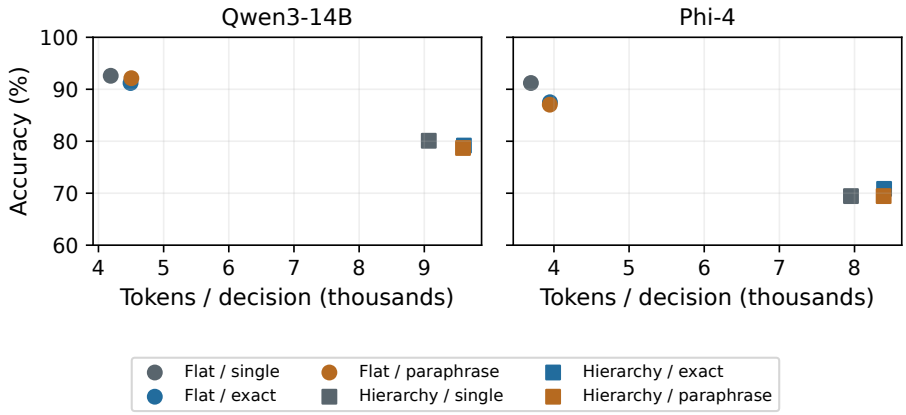


Fig. 5. Validation scheduling accuracy and total tokens per decision. Each arm contains 18 independent bases.

Declaration on AI Assistance

The human author supplied the research direction and experimental requirements. An AI coding assistant helped implement and test the benchmark, operate open-weight model experiments, retrieve literature, generate analyses, and draft the manuscript. This assistance extended beyond language editing. The author has not yet completed scientific review of the methods, interpretations, references, and conclusions in this working draft. Those contributions require author verification and revision before submission. No AI system is listed as an author.

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